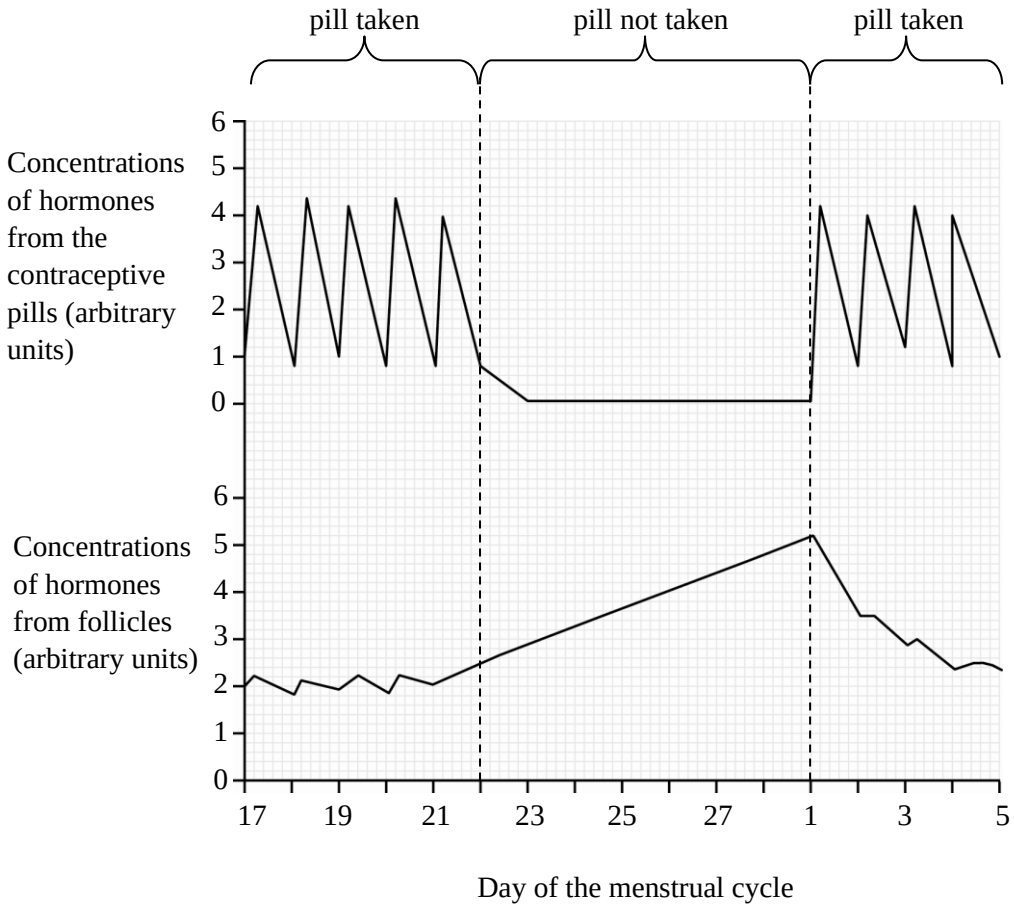
Structured questions

|!|BCELB0350000001|!| ★

Contraceptive pills use synthetic oestrogen and progesterone to achieve contraception. The following graph shows the blood concentrations of hormones released by contraceptive pills and ovarian follicles during a course of such pills.



(a) Explain the illustrated change in the blood concentration of hormones from the contraceptive pill. (3 marks)

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(b) Given the graph above, describe and explain how the course of contraceptive pills affects the blood concentration of hormones secreted by the follicles

(i) between day 17 and day 22; and (3 marks)

(ii) between day 22 of one menstrual cycle and day 1 of the next cycle. (3 marks)

(c) Although contraceptive pills can prevent pregnancies effectively, their use of oestrogen and progesterone can cause physiological issues. Give one example. (1 mark)

##

(a) The hormone concentration rises when the pill is taken and its hormones are absorbed into the blood. (1)
The concentration then falls as the hormones are gradually broken down by the liver. (1)
The hormone concentration remains low in the period when no pills are taken. (1)

(b) (i) Between day 17 and day 22, the concentration of the hormones secreted by the follicles remains almost constant at a low level. (1)
This is because high levels of oestrogen and progesterone from the contraceptive pills inhibit the secretions of FSH and LH from the pituitary gland. (1)

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The low level of FSH is insufficient to stimulate follicle development and by extension the follicular secretion of hormones. (1)

(ii) Between day 22 and day 1, the concentration of hormones secreted by the follicles rises. (1)

This is because the oestrogen and progesterone from the contraceptive pills is now absent, thus the secretion of FSH is no longer inhibited. (1)

The pituitary gland secretes FSH, which stimulates the development of follicles and causes their secretion of oestrogen. (1)

(c) Any ***one*** of the following: (1)

- Contraceptive pills may cause unwanted side effects (e.g. irregular menstruation, nausea, etc.).
- Contraceptives that act through hormones can upset the natural hormone balance in the body.

(Or other reasonable answers)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)(i)	Analysing data	1
	Applying concepts	2
(b)(ii)	Analysing data	1
	Applying concepts	2
(c)	Understanding basic concepts	1

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|!|BCELB0350000002|!| ★

Figure I shows the change in the levels of two hormones during one menstrual cycle of a woman. **Figure II** shows the structures that produce these hormones.

Figure I

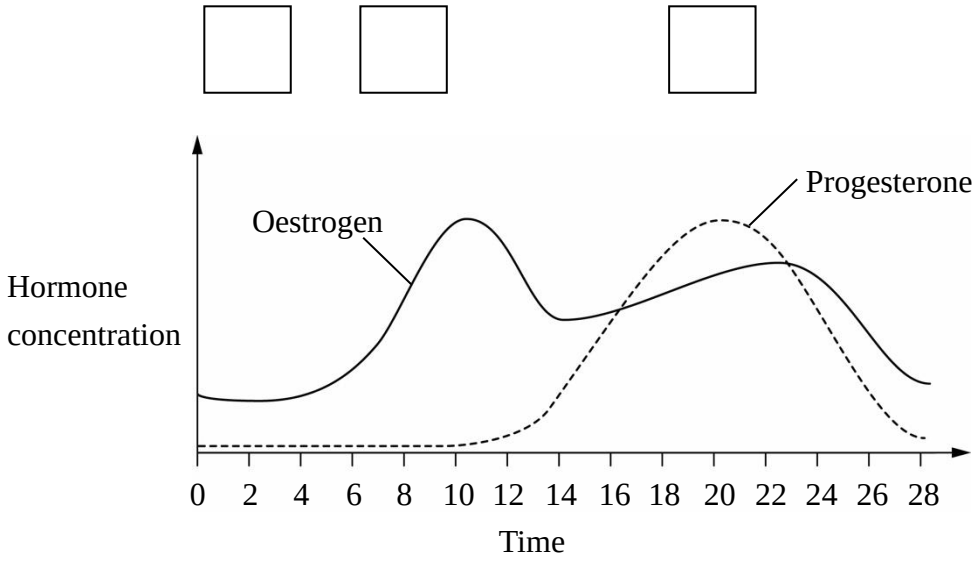
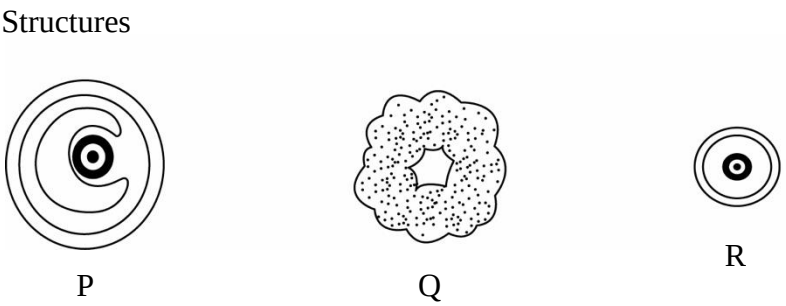


Figure II



(a) Fill in the boxes in Figure 1 with the correct letter labels to show the order in which the structures P, Q and R appear during the cycle. (1 mark)

(b) Explain how the level of oestrogen changes between day 0 and day 10. (4 marks)

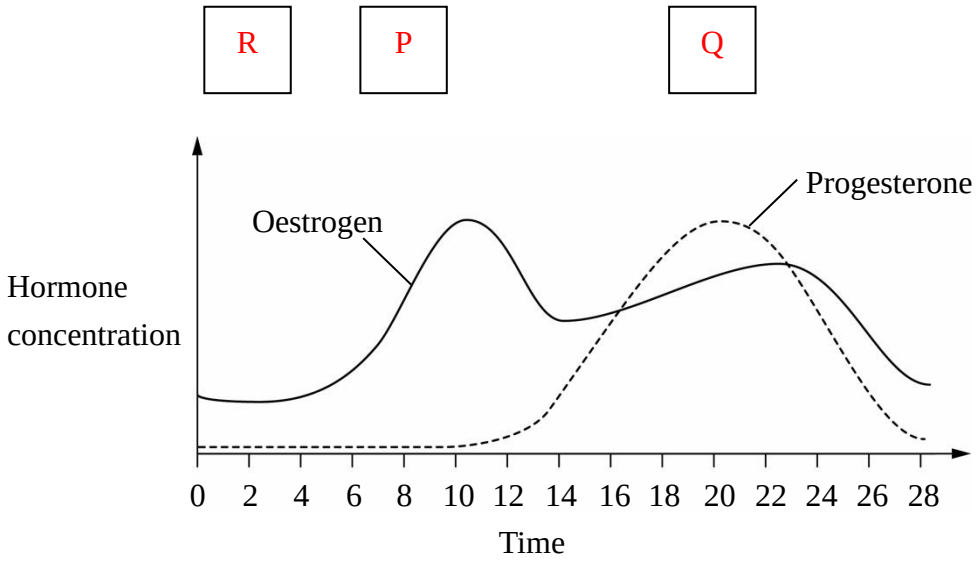
(c) Explain how the level of oestrogen changes on day 11. Then explain the subsequent increase in the levels of both oestrogen and progesterone. (3 marks)

(d) The levels of progesterone and oestrogen remain high after ovulation. Explain how this can prepare the body for pregnancy. (3 marks)

(e) Immediately before ovulation, a critical peak level of oestrogen stimulates the pituitary gland to secrete follicle stimulating hormone (FSH) and luteinising hormone (LH). State the term that describes this regulatory mechanism. (1 mark)

##

(a)



(1)

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(b) The pituitary gland secretes follicle stimulating hormone (FSH). (1)
FSH stimulates follicle development in the ovary. (1)
The developing follicle secretes oestrogen, (1)
thus increasing the level of oestrogen. (1)

(c) On Day 11, the level of oestrogen drops because the mature follicle ruptures during ovulation and stops secreting oestrogen. (1)
After ovulation, the ruptured follicle develops into a yellow body/corpus luteum, (1)
which secretes both oestrogen and progesterone, thus increasing their levels. (1)

(d) Progesterone and oestrogen maintain the thickness of the uterine lining / stimulate the development of blood vessels in the uterine lining in preparation for the implantation of an embryo. (1)
The high levels of progesterone and oestrogen also inhibit the secretion of FSH and luteinising hormone (LH). (1)
This prevents the development and ovulation of other follicles during this period in the cycle. (1)

(e) Positive feedback (1)

(12 marks)

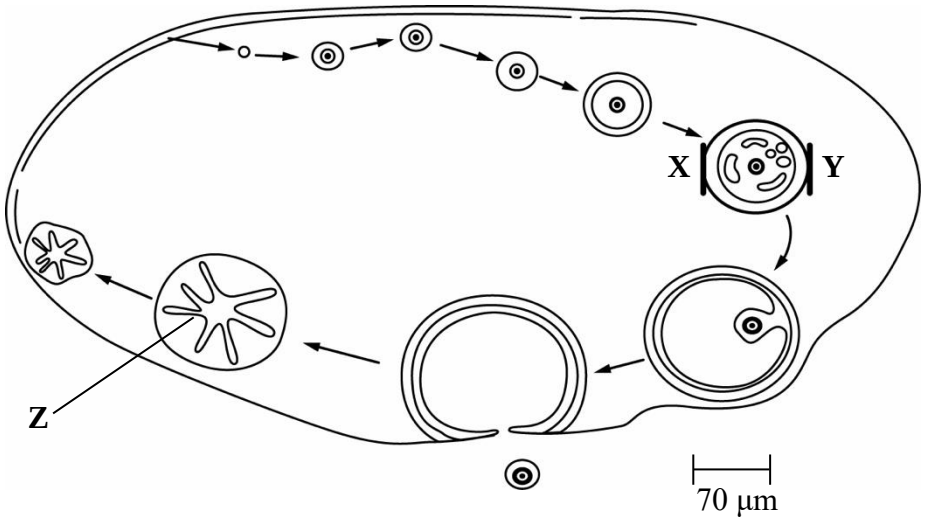
##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	4
(c)	Applying concepts	3
(d)	Understanding basic concepts	3
(e)	Understanding basic concepts	1

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|!|BCELB0350000003|!| ★ **STSE**

(a) The following diagram shows the developmental stages of one follicle in a human ovary.



(i) Calculate the actual length between X and Y on the ovarian follicle. Show your working. (2 marks)

(ii) Name structure Z. What will happen to this structure if fertilization does not occur? (2 marks)

(iii) Suggest why the menstrual cycle is coordinated by hormones and not by the nervous system. (3 marks)

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(b) Certain women who fail to conceive can take clomiphene, which blocks the action of oestrogen.

(i) Explain how treatment with clomiphene might cause ovulation. (4 marks)

(ii) What is the disadvantage of taking clomiphene to attain conception? (1 mark)

##

(a) (i) 70 μm × 1.4 (± 0.1) (1)
= 98 μm (Answer range: 91–105 μm) (1)
(ii) Corpus luteum / yellow body (1)
It will degenerate. (1)
(iii) Coordination of the menstrual cycle occurs over extended periods. (1)
Hormones can stay in the blood for a period of time and can thus exert long-lasting effects. (1)
In contrast, the nervous system's nerve impulses can only coordinate short-term / immediate responses. (1)

(b) (i) As oestrogen's action is blocked, it cannot inhibit the release of follicle stimulating hormone (FSH) and luteinising hormone (LH). (1)
FSH and LH can be secreted from the pituitary gland. (1)
FSH stimulates follicle development. (1)
After the follicle matures, LH can stimulate ovulation. (1)
(ii) Clomiphene can cause multiple ovulations and potentially multiple births. (1)

(12 marks)

##

	Skills assessed	Marks
(a)(i)	Mathematical skills	2
(a)(ii)	Analysing data	1
	Understanding basic concepts	1
(a)(iii)	Applying concepts	3
(b)(i)	Applying concepts	4
(b)(ii)	Applying concepts	1

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|!|BCELC0350000004|!| **STSE**

(a) Complete the table below to list ***one*** target tissue and ***one*** action for each of the two following hormones: luteinising hormone (LH) and progesterone. (4 marks)

Hormone	**Target tissue**	**Action during human menstrual cycle**
LH		
progesterone		

(b) Combined oral contraceptive pills can be taken by women to achieve contraception.

(i) Which ***two*** hormones do combined oral contraceptive pills contain? (2 marks)

_____

_____

(ii) How do combined oral contraceptive pills achieve contraception? (3 marks)

_____

_____

_____

(c) State ***two*** possible social implications of using contraceptives. (2 marks)

_____

_____

##

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(a) 1 mark for each box (4)

Hormone	Target tissue	Action during human menstrual cycle
LH	Follicle / yellow body	Any ***one*** of the following: • stimulates ovulation • stimulates the formation of the corpus luteum / yellow body in the ovary • stimulates the secretion of oestrogen and progesterone by the yellow body • prevents the degeneration of the yellow body
progesterone	uterine lining / the pituitary gland	Any ***one*** of the following: • maintains the thickness of the uterine lining • together with oestrogen, inhibits the secretion of follicle-stimulating hormone (FSH) and LH by the pituitary gland

(b) (i) Oestrogen (1)

Progesterone (1)

(ii) The synthetic oestrogen and progesterone in the pills inhibit FSH and LH secretions from the pituitary gland. (1)

This prevents follicle development and ovulation. (1)

As a result, no ovum is released and no fertilization can take place. (1)

(c) Any ***two*** of the following: (2)

- Fewer people suffer from poverty / starvation / diseases.
- Parents can provide more care for their children.
- More women can work.
- Increased promiscuity causing a greater incidence of sexually transmitted disease.
- Decreased population size leading to a smaller labour force in the future.

(Or other reasonable answers)

(11 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	4
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	3
(c)	Applying concepts	2

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|!|BCELA0350000005|!| ★★

Follicle stimulating hormone (FSH) and luteinising hormone (LH) are both composed of two polypeptide subunits: the α and β subunits. FSH and LH have identical α subunits while their β subunits differ. The β subunit of FSH has 111 amino acids and that of LH has 121 amino acids.

FSH and LH bind to different receptors in the cell membrane of their target cells. Such binding stimulates the synthesis of steroid hormones.

(a) Explain why LH does not bind to a FSH receptor. (3 marks)

(b) State which parts of the human ovary contain cells that possess

(i) FSH receptors (1 mark)

(ii) LH receptors (1 mark)

(c) Describe how the binding of FSH to the receptors on its target cells will affect the uterine lining. (3 marks)

(d) Name the hormone(s) that are produced as a result of the binding of LH to the receptors in its target cells. (1 mark)

##

(a) The β subunit of LH has a longer / different amino acid sequence than FSH, (1)
so both hormones have different conformation / tertiary structures. (1)
Thus the structure of LH is not complementary to the binding site of a FSH receptor and the hormone cannot bind to the receptor. (1)

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(b) (i) Follicle (1)

(ii) Corpus luteum / yellow body / follicle (1)

(c) The binding of FSH to its receptors in follicular cells (the target cells of FSH) stimulates follicle development. (1)

More oestrogen is secreted by the developing follicle. (1)

Oestrogen stimulates the thickening of the uterine lining. (1)

(d) Oestrogen and progesterone (1)

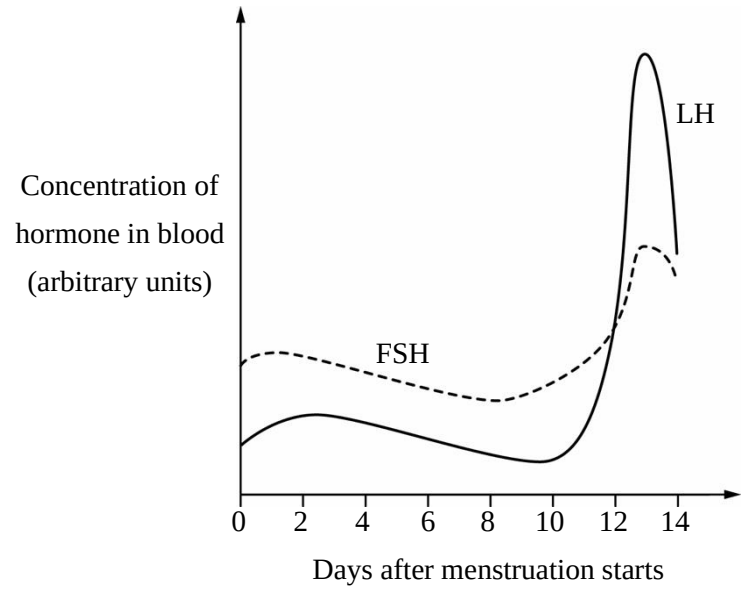
(9 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	1
(c)	Applying concepts	3
(d)	Applying concepts	1

|!|BCELB0350000006|!| ★

The graph below shows the changes in the blood concentrations of follicle stimulating hormone (FSH) and luteinising hormone (LH) during the first half of the menstrual cycle.



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(a) For the first half of the menstrual cycle,

(i) state the effect of FSH on the ovary. (1 mark)

(ii) state the effect of LH on the ovary. (1 mark)

(b) What causes the increase in the concentrations of FSH and LH between day 9 and day 13, as shown in the graph? (2 marks)

(c) What causes the decrease in the concentrations of FSH and LH on day 13? (2 marks)

(d) In the preparation for *in vitro* fertilization (IVF), women are injected with FSH. Explain why this treatment is necessary. (2 marks)

##

(a) (i) FSH stimulates follicle development. (1)

(ii) LH stimulates ovulation / the rupture of a mature follicle, releasing an ovum. (1)

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(b) A developing follicle secretes oestrogen, thus raising the concentration of oestrogen in the blood. (1)
Eventually a critical peak concentration of oestrogen is reached, stimulating the pituitary gland to secrete more FSH and LH into the blood. (1)

(c) After ovulation on day 13, the ruptured follicle develops into the yellow body / corpus luteum, which secretes oestrogen and progesterone. (1)
High concentrations of oestrogen and progesterone inhibit the release of FSH and LH by the pituitary gland. (1)

(d) The additional FSH can stimulate the ovary to produce many mature ova at the same time. (1)
IVF has a low success rate, thus it requires the harvest of many ova. (1)

(8 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Applying concepts	2
(d)	Applying concepts	2

|!|BCELB0350000007|!| ★

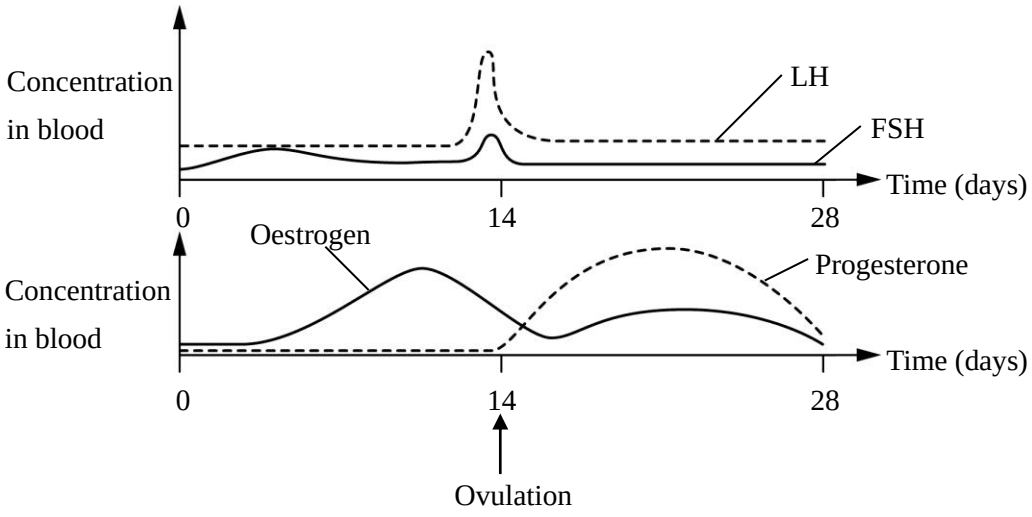
(a) State the site of production for the following hormones in human females:

(i) follicle stimulating hormone (FSH) (1 mark)

(ii) progesterone (1 mark)

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(b) The graph below shows the changes in the concentration of four hormones in the blood of a human female over 28 days. The four hormones are follicle stimulating hormone (FSH), luteinising hormone (LH), oestrogen and progesterone.



(i) Explain the relationship between the rise in the blood concentration of FSH from day 0 to day 4 and ovulation. (4 marks)

(ii) Explain how the changing concentrations of oestrogen and progesterone in the blood affect the uterine lining during the menstrual cycle. (3 marks)

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(c) The table below shows the mean size of human ovaries at different ages.

Age (years)	Mean size of human ovaries (cm^3)
8	0.5
10	0.7
12	2.2
14	3.1
16	4.0

(i) Using the tabulated data above, calculate the mean growth rate for ovaries between ages 12 and 16 years. Show your working. (2 marks)

(ii) Explain how the development of the ovaries during puberty will alter the appearance of females. Give ***two*** examples of such changes. (3 marks)

##

(a) (i) Pituitary gland (1)

(ii) Corpus luteum / yellow body (1)

(b) (i) The increased concentration of FSH stimulates follicle development in the ovary. (1)

The developing follicle secretes oestrogen. (1)

A peak in the blood concentration of oestrogen stimulates the pituitary gland to release LH. (1)

LH triggers ovulation. (1)

(ii) The rise / peak in oestrogen concentration in the blood (before ovulation / day 14) stimulates the thickening of the uterine lining. (1)

The rise / peak in progesterone concentration in the blood (after ovulation / day 14) maintains the thickness of the uterine lining. (1)

A drop in progesterone concentration in the blood (after day 21) causes the uterine lining to break down. (1)

(c) (i) $(4 – 2.2)\ cm^3$ / (16 – 12) years (1)

$= 0.45\ cm^3$ / year (1)

(ii) Developing ovaries will secrete oestrogen, which causes the development of secondary sexual characteristics. (1)

These include (any ***two*** of the following): (2)

- enlargement of the breasts

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- growth of pubic hair
- widening of the hips
- accumulation of subcutaneous fat deposits

(Or other reasonable answers)

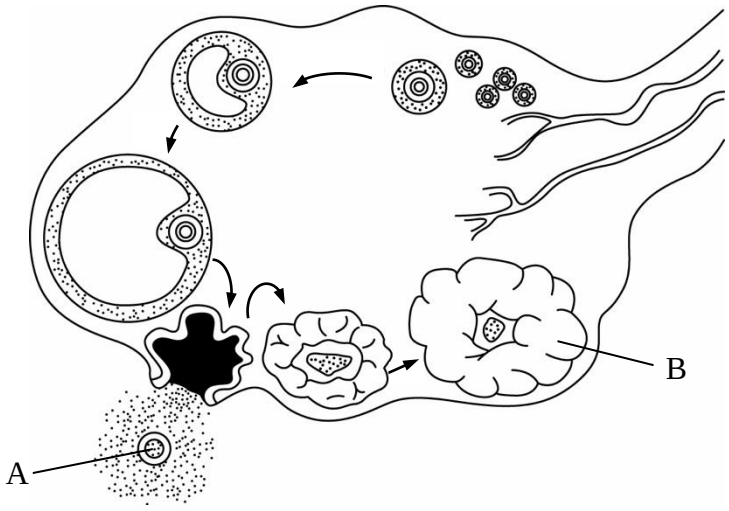
(14 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(b)(i)	Applying concepts	4
(b)(ii)	Applying concepts	3
(c)(i)	Mathematical skills	2
(c)(ii)	Understanding basic concepts	3

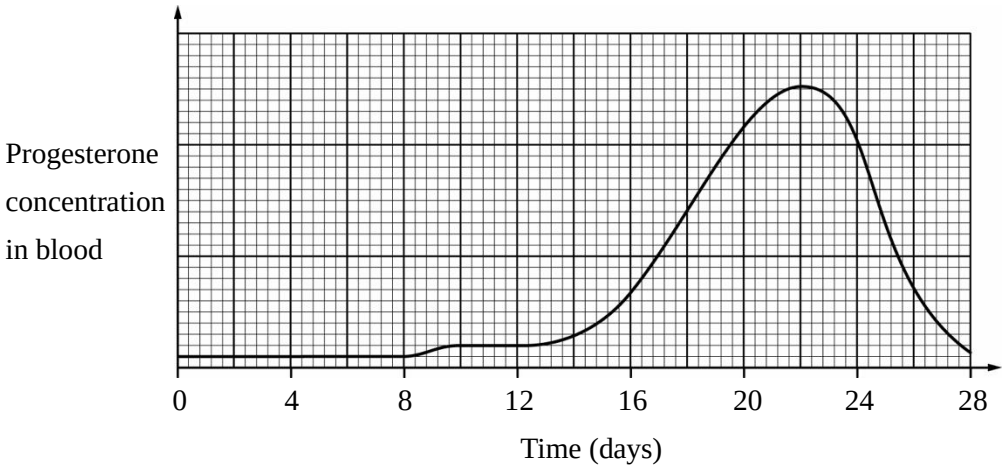
|!|BCELA0350000008|!| ★★

The diagram below shows the sequence of changes that a follicle undergoes within an ovary during one menstrual cycle.



(a) Name the structures labelled A and B. (2 marks)

(b) The graph below shows the changes in the concentration of progesterone in the blood during the menstrual cycle.



(i) Explain the drop in the progesterone concentration. (2 marks)

(ii) If fertilization occurs on day 14, the concentration of progesterone in the blood changes from that shown in the graph.
On the graph above, draw a curve to show this change after day 14. (2 marks)

(c) The table below shows the changes in the concentration of progesterone in the blood during the first 36 weeks of pregnancy.

Time (weeks)	0	4	8	16	24	32
Concentration of progesterone (arbitrary units)	7	8	9	11	15	30

(i) Describe the changes in the concentration of progesterone as shown in the table. (2 marks)

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(ii) Give ***two*** functions of progesterone in the menstrual cycle. (2 marks)

_____

_____

##

(a) A: ovum (1)

B: corpus luteum / yellow body (1)

(b) (i) The yellow body degenerates when fertilization does not occur. (1)

The degenerated yellow body stops secreting progesterone. (1)

(ii) Draw a curve from the peak and extend it towards the right hand side (from day 22 to day 28) (1)

The curve remains at the same level as the peak / rises as it progresses towards the right hand side. (1)

(c) (i) The concentration of progesterone increases steadily in the first 24 weeks. (1)

The concentration of progesterone increases sharply after 24 weeks. (1)

(ii) Any ***two*** of the following: (2)

- Together with oestrogen, it inhibits FSH / LH secretion.
- It maintains the thickness of the uterine lining.
- It inhibits ovulation.
- It thickens the mucus in the cervix.

_____

(10 marks)

##

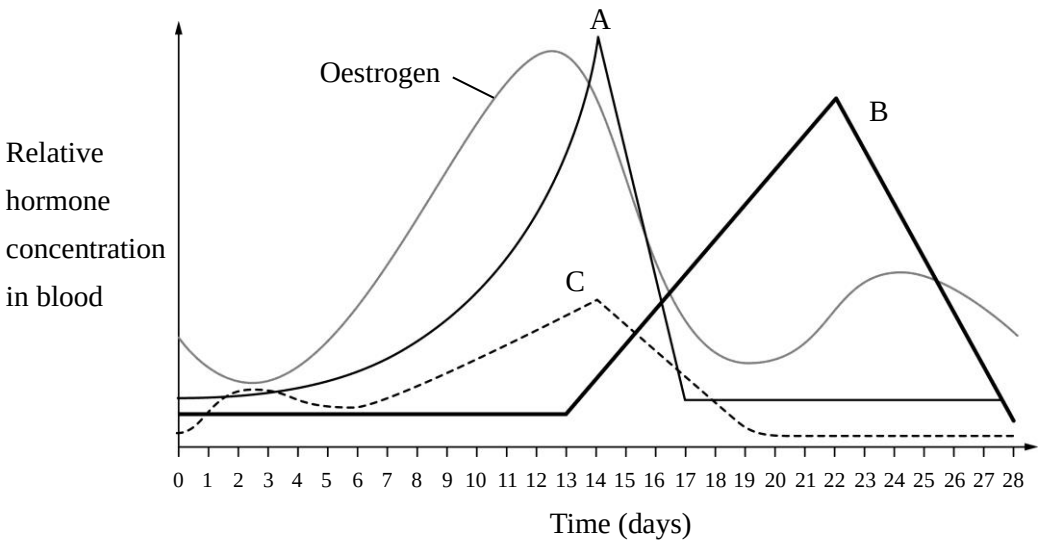
	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	2
(c)(i)	Analysing data	2
(c)(ii)	Understanding basic concepts	2

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|!|BCELA0350000009|!| ★★

(a) **Figure 1** shows the relative concentrations of follicle stimulating hormone (FSH), luteinising hormone (LH), progesterone and oestrogen in the blood of a non-pregnant female during one menstrual cycle.

Figure 1



Identify the hormones A, B and C in **Figure 1** and state ***one*** of their respective functions in the menstrual cycle. (6 marks)

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(b) According to research, about three quarters of all adult asthma patients are women. Researchers have suggested that variation in the blood concentration of oestrogen during the menstrual cycle may be a risk factor for acute asthma attacks in women. They collected data from 190 non-pregnant women under 47 years of age about when their last acute asthma attacks occurred. The collected data are shown below in **Table 1**.

Table 1

Phase of menstrual cycle (days)	Number of women with acute asthma attacks
5 – 11	38
12 – 18	45
19 – 25	20
26 – 4	87

(i) Calculate the percentage of women who had an acute asthma attack between day 26 of one menstrual cycle and day 4 of the next. Show your working. (2 marks)

(ii) Describe the results shown in **Table 1**. (2 marks)

(iii) With reference to **Figure 1** and **Table 1**, state the supporting evidence for the suggestion that variation in the blood concentration of oestrogen is a risk factor of acute asthma attacks. (2 marks)

##

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(a) A is luteinising hormone (LH). (1)

Any ***one*** of the following functions of LH: (1)

- Its presence in high concentrations (on around day 14) stimulates ovulation.
- After ovulation, it causes the ruptured follicle to develop into a yellow body / corpus luteum.

B is progesterone. (1)

Any ***one*** of the following functions of progesterone: (1)

- It maintains the thickness of the uterine lining.
- It thickens the mucus in the cervix.
- Together with oestrogen, it inhibits the secretion of FSH / LH and thus also inhibits ovulation.

C is follicle stimulating hormone (FSH). (1)

FSH stimulates follicle development in the ovary. (1)

(b) (i) (87 / 190) × 100% (1)

= 46% (1)

(ii) The number of acute asthma attacks is the highest between day 26 and day 4 / rises between day 5 and day 18. (1)

The number of acute asthma attacks is the lowest between day 19 and day 25 of the cycle. (1)

(iii) The chance of acute asthma attacks is lower when oestrogen concentration is high. (e.g. day 5 – 11, day 19 – 25) (1)

The chance of acute asthma attacks is higher when oestrogen concentration is decreasing. (e.g. day 12 – 18, day 26 – 4) (1)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
	Understanding basic concepts	3
(b)(i)	Mathematical skills	2
(b)(ii)	Analysing data	2
(b)(iii)	Analysing data	2

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|!|BCELC0350000010|!|

(a) State and explain the changes that occur in the uterine lining in response to varying hormone levels during the different phases of the menstrual cycle. (6 marks)

(b) List ***three*** conditions that will alter the menstrual cycle in adult females. (3 marks)

##

(a) During menstruation, the uterine lining breaks down (1)
in response to low levels of progesterone and oestrogen in the blood. (1)
During follicle development, the uterine lining thickens (1)
due to the increased level of oestrogen secreted by the developing follicle. (1)
After ovulation, the thickness of the uterine lining is maintained (1)
by the progesterone secreted by the yellow body / corpus luteum. (1)

(b) Any ***three*** of the following: (3)

- pregnancy / implantation
- taking contraceptive pills / receiving contraceptive injections / using contraceptive patches
- taking fertility drugs that contain hormones
- menopause
- ovarian cancer
- lactation / breast feeding

(Or other reasonable answers)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	6
(b)	Understanding basic concepts	3

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|!|BCELC0350000011|!| **STSE**

Women with infertility cannot achieve pregnancy naturally, but they can be treated with *in vitro* fertilization (IVF) and hormonal fertility drugs.

(a) Suggest ***two*** possible objections to the use of IVF. (2 marks)

(b) Give ***two*** common causes of infertility that arise in the female reproductive system. (2 marks)

(c) Explain how hormonal fertility drugs can overcome certain common causes of female infertility. (4 marks)

##

(a) Any ***two*** of the following: (2)

- Unused embryos may be destroyed.
- It is an unnatural process.
- It goes against religious beliefs.
- It is costly.
- It has a low success rate.
- It is more likely to cause multiple births.
- It raises controversies related to selective abortion / eugenics.

(Or other reasonable answers)

(b) Any ***two*** of the following: (2)

- Ovary fails to develop mature follicles / ovulate
- Embryo fails to implant into the uterine lining
- Oviduct / fallopian tube blockage

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(Or other reasonable answers)

(c) Fertility drugs containing FSH and LH can be used to treat infertility that is caused by problems with ovulation. (1)

The additional FSH and LH in the drug further promote follicle development and stimulate ovulation, respectively. (1)

Fertility drugs containing progesterone can be used to treat infertility that is caused by problems with implantation. (1)

Progesterone maintains the thickness of the uterine lining to facilitate the possible implantation of an embryo. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	4

|!|BCELB0350000012|!| ★ **STSE**

(a) Women who do not ovulate are infertile, although they may be treated with daily injections of follicle stimulating hormone (FSH). The concentration of oestrogen in the women's blood will be measured every day. When the oestrogen rises to a certain concentration, a subsequent injection of luteinising hormone (LH) will be given.

(i) Explain why the concentration of oestrogen rises in women who have received FSH injections. (2 marks)

(ii) Explain why LH is given to women as part of the treatment of infertility. (1 mark)

(b) Menopause is the time in a woman's life when her menstrual periods stop permanently and she is no longer fertile. By this time, menopausal women will have few follicles left in their ovaries. Explain why these women have high levels of FSH in their blood. (3 marks)

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(c) Some women may take contraceptives that contain progesterone in order to prevent pregnancy after sexual intercourse. State and explain ***two*** ways through which progesterone acts as a contraceptive. (4 marks)

##

(a) (i) FSH stimulates follicle development in the ovary. (1)
Developing follicles secrete oestrogen into the blood. (1)
(ii) LH can stimulate ovulation to occur / the mature follicle to rupture and release the ovum inside. (1)

(b) Oestrogen is not produced / Little oestrogen is produced because there are no / few follicles in the ovaries. (1)
The low level of oestrogen no longer inhibits the pituitary gland from secreting FSH. (1)
Less FSH is consumed due to the fewer number of follicles, thus its level in the blood increases greatly. (1)

(c) Any ***two*** sets of the following: (4)

- A high level of progesterone inhibits the secretions of FSH and LH. (1)
  Low levels of FSH and LH prevent follicle development and ovulation, respectively. (1)
- Progesterone causes the thickening of the cervical mucus. (1)
  This makes it difficult for sperm to pass through the cervix. (1)
- Progesterone prevents follicle development and the subsequent release of oestrogen from the follicles. (1)
  This prevents the uterine lining from thickening and renders it unable to receive embryo implantation. (1)

(10 marks)

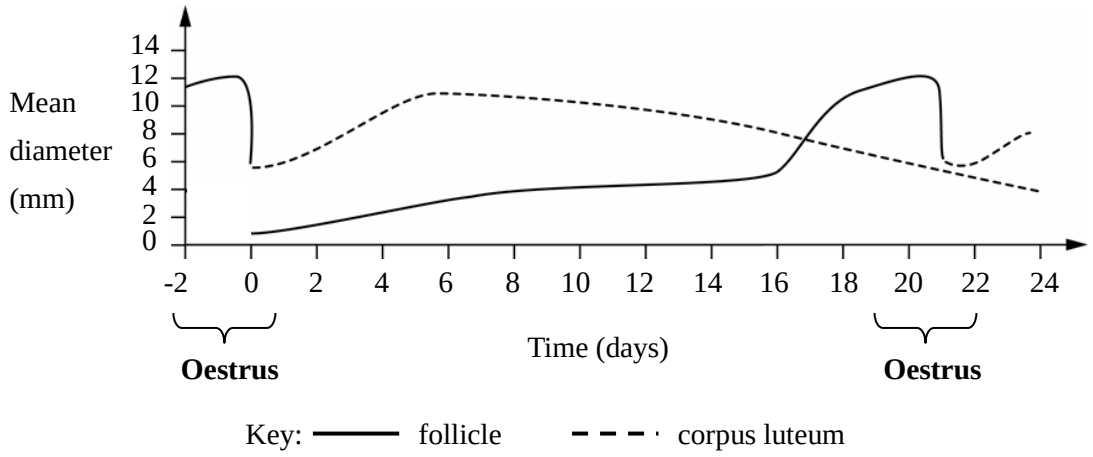
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	Skills assessed	Marks
(a)(i)	Applying concepts	2
(a)(ii)	Applying concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	4

|!|BCELA0350000013|!| ★★ **STSE**

The graph below shows the changes in the diameters of the follicle and the corpus luteum during the oestrus cycle of a pig. The oestrus cycle is defined as the time period between successive ovulations. Oestrus is the period in which a female pig is sexually receptive.



(a) In this oestrus cycle, ovulation occurs on day 21. Explain how the graph gives evidence for this. (2 marks)

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The table below shows the variation in progesterone and oestrogen concentrations during the oestrus cycle of the pig.

Time (days)	Concentration of hormone in the blood (arbitrary units)	
	Progesterone	**Oestrogen**
0	1.8	1.0
2	4.7	0.9
4	9.0	0.9
6	12.1	0.9
8	12.0	0.9
10	8.7	0.9
12	3.5	0.9
14	0.9	1.7
16	0.6	2.1
18	0.6	6.0
20	1.1	3.2
22	3.7	0.9
24	7.5	0.9

(b) Using the data from the graph and table above, describe and explain the relationship between the diameter of the corpus luteum and the concentration of progesterone in the pig's blood between day 0 and day 14. (2 marks)

(c) Use the data from the table above and your own knowledge to explain how the changing concentrations of oestrogen and progesterone regulate the oestrus cycle. (6 marks)

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(d) In an investigation, synthetic progesterone was added to the diet of female pigs for about 14 days. The female pigs then all came into oestrus 5 days later.

(i) Give ***one*** reason why it is useful for a farmer to be able to determine when oestrus will occur? (1 mark)

(ii) However, there is public opinion against feeding pigs with reproductive hormones. Give ***one*** biological argument that can support this opinion. (1 mark)

##

(a) Follicle diameter / size drops suddenly on day 21. (1)
A corpus luteum is formed on day 21. (1)

(b) The concentration of progesterone in the pig's blood increases along with the diameter of the corpus luteum. (1)
This is because the corpus luteum secretes progesterone. (1)

(c) Progesterone inhibits the secretions of FSH and LH. (1)
Once progesterone concentration falls (on day 16), FSH increases in concentration. (1)
FSH stimulates follicle development. (1)
Developing follicles secrete oestrogen. (1)
The peak oestrogen level on day 18 stimulates the secretion of FSH and LH. (1)
A high level of LH stimulates ovulation and then the formation of the corpus luteum. (1)

(d) (i) Any ***one*** of the following: (1)
- The farmer can know when to perform artificial insemination.
- The farmer can know when to introduce male pigs to mate with the female pigs.
- The farmer can ensure synchronized birthing.

(Or other reasonable answers)

(ii) Any ***one*** of the following: (1)
- The synthetic hormones may be present in the animal's meat / ingested by humans.
- The synthetic hormones may be toxic / cause undesirable effects in humans / affect the human female menstrual cycle / alter fertility.
- There is no need to apply the hormones because a pig's oestrus pattern can be predicted via behaviour / urine analysis.

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(Or other reasonable answers)

(12 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	2
(c)	Applying concepts	6
(d)(i)	Applying concepts	1
(d)(ii)	Critical thinking	1

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